Technical review

Item #2 · 12-mailbox.mbox#2@1372 · Run ID: 20261113T232800Z-08b8d1

Path	`R:\evidence\mail\12-mailbox.mbox#2@1372`
Size (bytes)	1276
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.mbox
Detected type (signature)	Email message in an mbox (RFC 5322)
Status	Problem found
Derived from (mbox evidence)	Message #2 of the mbox R:\evidence\mail\12-mailbox.mbox (md5 f4e8e05ccc50d543ed3703f21cb965d4, sha256 2e39de637dad0f16469632facf7544dfe4fc0d39bf7210dd579b7a74ba0f5ebe), stored from byte offset 1421, 1276 bytes (separator line at offset 1372). The custody of these bytes is the custody of that mbox file.
Case	RV-SAMPLE-001
Evidence ID	E0016
Case registry	
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`a62b0d52faaaad62074e6f84f0f9c08c998c11138514f3f9db5a459231ab5c50`
md5	`a3bd09e3d5582c8dc5c000d71e455d3f`

Findings

[High] DKIM body hash does not match (DKIM_BODY_HASH_MISMATCH)

What it shows	The DKIM signature of example.org (selector s2026) records a hash of the message body that differs from the hash of the body as it is now: the body is not the one that was signed.
What it does not prove	It does not show who changed the body, when, or why: forwarding systems, mailing lists, antivirus footers and re-encoding by a mail client also change bodies after signing.

Email analysis

Message in mbox	#2 @1372 (1276 B): From bruno@example.org Wed Nov 11 11:30:02 2026
Message-ID	20261111123000.13@example.org
From	Bruno Proveedor <bruno@example.org>
Return-Path	bruno@example.org
Subject	Albarán 32
Date header	Wed, 11 Nov 2026 12:30:00 +0100 (2026-11-11T11:30:00Z)
DKIM mode / key file	offline · R:\inputs\dkim-keys.txt · 4 key(s)

Received hops (first server = 1)

Hop 1	From host: [192.0.2.10] · Received by: smtp.example.org · Date stamped: 2026-11-11T11:30:00Z

Hop 2	From host: smtp.example.org · Received by: mx.acme.example · Date stamped: 2026-11-11T11:30:02Z

DKIM signatures (offline verification)

#1 d=example.org s=s2026	Result: body_hash_mismatch · Algorithm: rsa-sha256 · Key source: key_file (line 3) · computed body hash differs from bh=
ARC chain	none (0)

Files written by the analysis

headers.json (headers)	md5 7fa92a925c05fd5c5bb78b20605d57fb · sha256 97cca6a18734c381404de57e694052363e11e5db222af818f650a6c47a55eb26
dkim.json (dkim)	md5 a0651b6344eb47e8810dc299e33e873e · sha256 5407171bff26a0f7eefd07fc1609b58cd2acd32f9c77eedff52f04719b21590a

Method

Each file was read once to compute its digests, then analysed as email: .eml as an RFC 5322 message, .msg through its MAPI properties, and each message of an mbox as an item of its own (each in a separate worker process with time and memory limits when worker isolation is on, as recorded in the audit log). The header section, Received hops and dates were read and compared; DKIM and ARC signatures were verified offline with the public keys the examiner supplied (no DNS query unless online mode was chosen); Authentication-Results were recorded as stated evidence; attachments were hashed; and BEC heuristics compared the addresses with the organisation's own domains. Every step and every file written is recorded in the hash-chained audit log.

Audit trail

Audit entries	#25 – #30
Log folder	`R:\case\runs\rv-mailauth\20261113T232800Z-08b8d1\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- DKIM results depend on the public keys supplied: a check is only meaningful with the key that was valid when the message was sent. Keys are rotated, so a current key can fail a genuine message and a missing key leaves the signature unchecked.
- A missing DKIM signature is common and proves nothing; a valid signature covers only the signed header fields and the body, not who wrote the message.
- Outlook .msg files do not keep the original message: the body is re-encoded (so DKIM cannot be checked) and the original header section is kept only when transport headers were stored.
- Lookalike-domain, Reply-To, display-name and payment-wording checks are heuristics with false positives and false negatives; the organisational domain comes from the bundled Public Suffix List, whose version is recorded in environment.json and may differ from the list in force when the message was sent.
- Received, Date and Authentication-Results headers are text that servers and clients wrote; unsigned headers can be added or edited by anyone who handled the file.
- In an mbox, body lines starting with "From " are escaped with '>'; un-escaping follows the selected variant (mboxrd by default) and cannot always restore the original line, which can affect DKIM body hashes.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.